RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS261TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the five important functions of management.	2	1	1
2.	Identify which statement represents vision and which represents mission: i. "To be a globally recognized brand known for innovation." ii. "We design and deliver innovative products to meet customer needs."	2	1	1
3.	Do you agree that coordination need not be treated as a separate function of management? Give a brief reason.	2	2	1
4.	Explain how economists respond to the criticism that people often make irrational decisions due to emotions, habits, or lack of information.	2	2	3
5.	Explain how the government is involved in the circular flow of income.	2	2	3
	Part – B			
1	A manager at Swiggy handles team coordination, information sharing, and decision-making during peak hours. Apply Henry Mintzberg's managerial roles to illustrate how the manager performs these functions in this situation.	10	L3	1
2	In the launch of a new electric vehicle by Tata Motors, the project manager must coordinate teams, solve problems, and motivate employees. Apply the important managerial skills to show how the manager can successfully handle this project.	10	L3	1
3	Using a real-life example, explain the concept of circular flow of income to show the role of households, firms, government, and foreign sector.	10	L2	3
4	Explain the basic economic problem of scarcity and choice and describe how individuals and firms allocate limited resources.	10	L2	3
5	Describe the different types of economic systems and explain any two with examples.	10	L2	3

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Date : 13/04/2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part–A			
1.	State the objective of a supply chain.	1	1	CO1
2.	List the typical stages in a supply chain.	2	1	CO1
3.	_____ cycle occurs between the manufacturer and the supplier.	1	1	CO1
4.	Write the full form of the following supply chain macro processes: CRM, ISCM, SRM	2	2	CO1
5.	What is a company's supply chain strategy?	2	1	CO1
6.	Illustrate the relationship between number of facilities and total logistics cost.	2	2	CO2
	Part–B			
1	Explain the impact of supply chain decision phases on the overall profitability and success of manufacturing supply chains.	10	3	CO1
2	Illustrate and discuss cycle view of an automobile component manufacturing organization.	10	3	CO1
3	List the three basic steps to achieving this strategic fit. Evaluate the first step.	10	2	CO1
4	Unipart, a manufacturer of auto parts, is considering two B2B marketplaces to purchase its MRO supplies. Both marketplaces offer a full line of supplies at very similar prices for products and shipping. Both provide similar service levels and lead times. However, their fee structures are quite different. The first marketplace, Parts4u.com, sells all of its products with a 5 percent commission tacked on top of the price of the product (not including shipping). AllMRO.com's pricing is based on a subscription fee of $10 million that must be paid up front for a two-year period and a commission of 1 percent on each transaction's product price Unipart spends about $150 million on MRO supplies each year, although this varies with their utilization. Next year will likely be a strong year, in which high utilization will keep MRO spending at $150 million. However, there is a 25 percent chance that spending will drop by 10 percent. The second year, there is a 50 percent chance that the spending level will stay where it was in the first year and a 50 percent chance that it will drop by another 10 percent. Unipart uses a discount rate of 20 percent. Assume all costs are incurred at the beginning of each year (so Year 1 costs are incurred now and Year 2 costs are incurred in a year). Which B2B marketplace should Unipart buy its parts from?	20	3	CO2

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : 6th	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define the term ergonomics.	01	L_1	CO1
2.	Name any two benefits of ergonomics	01	L_2	CO1
3.	What is meant by human centred design?	01	L_1	CO1
4.	Mention one example of cognitive ergonomics while interfacing with websites?.	01	L_2	CO1
5.	List Give a practical example of a Functional Anthropometric Data.	01	L_2	CO2
6.	What are the two important landmark percentiles in the normal distribution are as applied to consideration of anthropometric variables in designs?	01	L_2	CO2
7.	State any one principle of workspace design for sitting tasks.	01	L_2	CO2
8.	What is the primary goal of anthropometry?	01	L_2	CO2
9.	Give an example of application of Newtonian anthropometrics in designs	01	L_3	CO2
10.	Expand the acronyms CTD.	01	L_1	CO2
	Part – B			
1.	Develop a step-by-step ergonomic evaluation plan for a computer based VDT based workstation. Your answer should include: Identification of key task elements, Application of anthropometric measurements, Postural analysis methods, Recommendations to reduce musculoskeletal risks.	10	L_4	CO1
2.	Distinguish between Macro ergonomics and Micro ergonomics. Outline the scope of work that is covered in both these areas of ergonomics. Give an example to justify the application of these specific to work systems design and equipment/workspace designs. List any two factors to be considered for each of these two types of ergonomics in designs.	10	L_2	CO1
3.	Consider the example of a handle Design of a Handheld Power Tool: How should you calculate the Grip diameter based on the 5th and 95th percentile values of a hand breadth. For easy - Thumb reach: where should the controls be placed considering natural thumb arc. Which force can be reduced by choosing - Material: such as soft rubberized textures? Which of the two grips whether inline grip or pistol grip must be so as to minimize wrist deviation. If you can reduce the handle mass to $\leq$ 1 kg and achieve the balance centre near grip, what is the implications for operator fatigue during the power tool operations? (Provide Sketches to highlight the engineering anthropometric design and to visualize the handles design of a Handheld power tool)	10	L_4	CO2

4.	A company is designing a sitting assembly station for both male and female operators in a watch assembly line. Apply anthropometric principles to determine optimal work surface height, Analyse how human variability influences design decisions, Discuss trade-offs between productivity, comfort, and injury prevention in this work context. (Sketch your ideas with clarity of critical design variables)	10	L_3	CO2
5a.	What is “dynamic sitting,” and why is it preferred over maintaining a “perfect” upright posture?	05	L_2	CO1
b.	The term Maximum Holding Time (MHT) is defined as The longest duration a specific physical posture or force can be maintained before failure or unacceptable discomfort. Give a practical example to emphasize as to why this variable is important for equipment/workspace design based on your own exposure and experience.	05	L_3	CO2

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Date : 16th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Human Resource Management & Analytics		**Course Code : IM364TA**

Note:
Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	HR manager advises a department head on hiring. What type of authority is this?	01	L1	CO1
2.	A company uses online interviews for hiring. Which trend is this in HRM?	01	L2	CO1
3.	Which HR group focuses on strategic decision-making?	01	L1	CO1
4.	What is talent management?	01	L1	CO1
5.	What are the key components of a job description?	01	L1	CO1
6.	What is HR forecasting?	01	L1	CO2
7.	List the three steps in Succession planning?	02	L2	CO2
8.	List any two internal sources of candidates.	01	L1	CO2
9.	Define Job analysis	01	L1	CO1
	Part – B			
1.	Explain the major trends shaping Human Resource Management in detail.	10	L2	CO1
2.	Imagine you are working in the HR department of a hospital called *CarePlus Medical Center*. The hospital wants to improve its hiring process for the role of Staff Nurse. Prepare: a) A Job Description for the role of Staff Nurse b) A Job Specification for the same role	10	L4	CO1
3.	A manufacturing company wants to analyze a job that involves repetitive manual tasks on the shop floor. Which job analysis methods would you recommend? Justify your answer by comparing alternative methods.	10	L3	CO1
4.	An e-commerce company is expanding and needs to hire for marketing, customer support, data analysis, and logistics roles. Identify different external sources for recruiting candidates and explain how each aligns with the company's needs.	10	L3	CO2
5.	Imagine you are the HR manager of a multinational company that is about to launch a recruitment campaign for a new product development team. The team will consist of software developers, project managers, and quality assurance specialists. a) Outline the steps of the recruitment process you would follow to attract and hire the best talent for this team. b) Explain how you would ensure that the process is efficient and cost-effective. Discuss any tools or technologies you might use to streamline this recruitment process.	10	L4	CO2

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DEPARTMENT OF
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Date : 06.11.2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS251TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Why do managers need to be both efficient and effective?	2	L1	01
2	Mention three important skills required to be an effective manager.	2	L2	02
3	Differentiate between Unity of command and Unity of direction.	2	L2	01
4	Why should managers have a basic knowledge of economics?	2	L2	04
5	What does the circular flow of income illustrate in an economy?	2	L1	04
	Part – B			
1	A manufacturing firm is facing issues with productivity and frequent machine breakdowns. The management decides to reorganize shop-floor activities and assign specific tasks to workers based on time and motion studies. Incentives are planned for those who achieve higher productivity. Analyse the above case using Frederick Taylor's Scientific Management principles. Which principles are being applied, and how could they help improve efficiency?	10	L2	01
2	A growing IT company notices that despite excellent pay, employees show low morale and poor teamwork. The management brings in a consultant who focuses on employee well-being, informal group interactions, and feedback sessions. Evaluate the management's approach using Elton Mayo's Human Relations Theory. What lessons can be learned from the Hawthorne Experiments in this context?	10	L3	02
3	A CEO represents the organization at a global conference, negotiates mergers, sets strategic goals, and ensures communication across all divisions. Explain how this situation reflects Mintzberg's 10 managerial roles, grouped into their respective categories.	10	L3	01
4	A leading automobile manufacturer notices that the demand for its vehicles has declined due to an increase in petrol prices and the entry of a new competitor offering electric vehicles at lower prices. Meanwhile, the government is implementing policies to stabilize inflation and boost employment. Using the principles of microeconomics and macroeconomics, analyse how both these levels of economic study help in understanding and addressing this situation.	10	L2	04
5	Explain the two-sector circular flow model with the help of an example. How does it demonstrate the interdependence between households and firms in an economy?	10	L2	04

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DEPARTMENT OF
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Date : 28th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles of Management and Economics		**Course Code : HS261TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Mention any two skills required by an effective manager.	2	L1	01
2	List any two principles of scientific management as proposed by F W Taylor.	2	L1	01
3	Highlight the importance of Hawthorne Studies.	2	L2	01
4	Differentiate between micro and macroeconomics.	2	L2	04
5	What are the two main sectors in a simple circular flow model?	2	L1	05
	Part – B			
1	A retail chain wants to optimize its store operations for better customer service and efficiency. How can the principles of Scientific Management and Administrative Theory help in this scenario?	10	L3	01
2	You are part of the organizing committee for a Techno-Cultural Fest at your institution. Apply the five primary functions of management—Planning, Organizing, Staffing, Directing, and Controlling—to successfully manage the event. Explain how each function can be implemented to ensure the fest runs smoothly and meets its objectives.	10	L3	01
3	A large multinational company is facing challenges in managing its various departments and adapting to changes in the external market environment. Apply **Systems Theory** and **Contingency Theory** to suggest how the company can improve its overall management and operations. Explain how these theories would help in managing the complexity and adapting to the changing conditions.	10	L4	01
4	Consider an economy that is experiencing high inflation and high unemployment simultaneously (stagflation). Using the Circular Flow Model of Economics, explain how inflation and unemployment are interconnected in this economy. How can government policies in the form of fiscal and monetary interventions help address these issues?	10	L4	04
5	A country is transitioning from a command economy to a market economy. Using an overview of Economic Systems, discuss the potential advantages and disadvantages of this shift.	10	L2	05

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Date : 28th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Supply Chain Management		**Course Code : IM362IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the processes under SRM – macro process.	2	1	CO1
2.	Sketch the implied uncertainty (demand and supply) spectrum and cite examples.	2	2	CO1
3.	Compare Efficient and Responsive Supply Chains(any two).	2	1	CO1
4.	Mention any two challenges to achieving and maintaining strategic fit.	2	1	CO1
5.	Illustrate the relationship between desired response time and number of facilities.	2	2	CO2
	Part – B			
1	Identify the five stages of a supply chain and sketch four process cycles. Explain the sub-processes in a supply chain process cycle of your choice for a manufacturing supply chain.	10	3	CO1
2	Illustrate and analyse the value chain for a manufacturing organisation.	10	3	CO1
3	Evaluate the various stages involved in expanding the scope of strategic fit for an automobile supply chain.	10	3	CO1
4	Discuss the role that information plays in the supply chain, as well as key information-related decisions that supply chain managers must make.	10	2	CO1
5	Select the supply chain of your choice evaluate the performance of a distribution network along two dimensions: Customer needs that are met (service); and Cost of meeting customer needs.	10	4	CO2

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Date : 29th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: ERGONOMICS		**Course Code : IM363IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define human factors engineering in one sentence.	01	L1	CO1
2.	Name any two objectives of ergonomics.	01	L2	CO1
3.	What is meant by user-centred design?	01	L2	CO2
4.	Mention one example of cognitive ergonomics in interface design.	01	L1	CO2
5.	List the three types of anthropometric data.	01	L1	CO2
6.	What does percentile mean in the context of anthropometric measurements?	01	L2	CO2
7.	State one principle of workspace design for sitting tasks.	01	L1	CO3
8.	What is the primary goal of ergonomics in design?	01	L1	CO2
9.	Differentiate between static and dynamic anthropometry with one word each.	01	L2	CO2
10.	Which part of the body is most commonly associated with low back pain due to poor workspace ergonomics?	01	L2	CO3
	Part – B			
1.	Develop a step-by-step ergonomic evaluation plan for a manufacturing workstation. Your answer should include: Identification of key task elements, Application of anthropometric measurements, Postural analysis methods, Recommendations to reduce musculoskeletal risks.	10	L3	CO1
2.	Analyze the causes of work-related musculoskeletal disorders (WMSDs) in the context of poor workspace design. Use the following structure: Ergonomic risk factors, Examples of bad design, Physiological consequences on the musculoskeletal system, Suggestions for redesign based on anthropometry.	10	L4	CO3
3.	A company is designing a standing workbench for both male and female operators in an electronics assembly line. Apply anthropometric principles to determine optimal workbench height, Analyze how human variability influences design decisions, Discuss trade-offs between productivity, comfort, and injury prevention.	10	L4	CO2
4.	The 97.5th percentile value and 2.5th percentile values of normally distributed variable is found by adding/subtracting 1.96 standard deviation from the mean. If the mean body mass of males of a certain populations is 82.1 Kg and the standard deviation is 17.1 kg, what are the 2.5th percentile value and the 97.5th percentile values? How will use the results of this calculation be used in considering the range of users covered in design?	10	L4	CO2

5.	Discuss the scope of work that is covered under the following type of ergonomic studies. i. Physical ergonomics. ii. Cognitive ergonomics. iii. Organizational ergonomics. iv. Macro ergonomics.	10	L2	CO1

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Date : 29th April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Human Resource Management & Analytics - IM364TA		

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is Talent Management?	02	L1	CO1
2.	List the type of information that can be collected through job analysis.	02	L2	CO1
3.	List the three steps in Succession planning.	02	L2	CO1
4.	Differentiate between Job specification and Job description.	02	L1	CO1
5.	Name the main internal sources of candidates.	02	L2	CO2
	Part – B			
1	Bright Minds School hired a new HR Manager, Ms. Riya, to transform its HR practices. While she has excellent knowledge of HR policies, she faces challenges in communication, conflict resolution, and strategic thinking. The management is unsure if she is the right fit for a role that requires leadership and organizational development. **Questions:** 1. Based on this case, identify which HR competencies are lacking and how these gaps can affect HRM outcomes. 2. Suggest ways Ms. Riya can develop the necessary competencies to succeed in her role. 3. Discuss how aligning HR objectives with organizational goals can enhance overall institutional effectiveness in this context.	10	L4	CO1
2	Imagine you are working in the HR department of a delivery company called SwiftCart Logistics. The company needs a clear job profile for the role of Delivery Executive to improve hiring and performance. Write the a) Job Description for the role of Delivery Executive and b) Job Specification for the same role.	10	L3	CO1
3	Name any four methods used for collecting Job Analysis information for roles like cashier.	10	L3	CO1
4	Mention two internal and two external sources of recruitment for roles like chefs. Provide one example for each.	10	L3	CO2

5	Ritz Resort is expanding its operations and requires new hires for key positions such as spa therapists, chefs, and front desk receptionists. The HR department is in charge of managing the entire recruitment process to ensure the right candidates are selected quickly and effectively. Questions: 1. Describe the key steps in the recruitment process that the HR team at Ritz Resort should follow when hiring spa therapists and chefs. 2. How can they ensure the quality of candidates selected for these roles?	10	L4	CO2

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**25th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**HS251TA**	**Duration**	**120 Min**
Sem	**V**	**CIE – I**	
PRINCIPLES OF MANAGEMENT AND ECONOMICS			

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Which step in POSDCORB involves delegating tasks and giving structured instructions?	01	1	1
2	Define "effectiveness" in the context of management functions.	01	1	1
3	Which management approach is most associated with using data and mathematical models to improve decision-making?	01	1	1
4	In the Contingency Approach, what determines the best management style?	01	1	2
5	_____ is the primary scientist associated with the foundation of Administrative Theory.	01	1	1
6	How Total Quality Management (TQM) incorporates the Quantitative approach to improve customer satisfaction.	01	1	2
7	What role does fiscal policy play in stabilizing an economy during a recession?	01	1	4
8	What is the role of Government within the circular flow model?	01	1	4
9	What are the key components of macroeconomics that measure the health of an economy?	02	1	5
	Part – B			
1a	Mention the underlying reasons for the increase in productivity during the Hawthorne Studies, and how do these reasons challenge the idea that only physical conditions affect workplace efficiency?	06	4	1
1b	How does Contingency Theory explain the limitations of a 'one-size-fits-all' approach to management? Provide two examples to illustrate how it applies to different organizational situations.	04	3	2
2	Compare and contrast the Classical and Behavioral approaches to management. How do these approaches address productivity, and what are their limitations in modern workplaces?	10	2	1
3	Analyze how POSDCORB ensures both efficiency and effectiveness in management. Discuss the potential challenges managers face if one of these functions is underperformed.	10	3	1
4	Evaluate the relevance of Systems Theory in managing modern organizations. How does its emphasis on interdependence and the environment benefit decision-making?	10	4	1

5	Analyze how microeconomics, macroeconomics, and economic systems interact within the framework of the circular flow model. What challenges arise when these components fail to work in harmony?	10	4	5

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max	Quiz	04	02	–	02	02	10	–	–	–	–	–
	Marks	Test	36	04	–	0	10	–	10	14	26	–	–